60. Use Cauchy's residue theorem to evaluate the following:

  (i) $\int_C \frac{(z-3)\mathrm{d}z}{z^2+2z+5}$, where C is the circle $|z+1-i|=2$

  (ii) $\int_C \frac{\mathrm{d}z}{(z^2+4)^3}$, where C is the circle $|z-i|=2$

  (iii) $\int_C \frac{(z-1)\mathrm{d}z}{(z+1)^2(z-2)}$, where C is the circle $|z+i|=2$.

4.5 CONTOUR INTEGRATION—EVALUATION OF REAL INTEGRALS

The evaluation of certain types of real definite integrals can be done by expressing them in terms of integrals of complex functions over suitable closed paths or contours and applying Cauchy's residue theorem. This process of evaluation of definite integrals is known as contour integration.

We shall consider only three types of definite integrals which are commonly used in applications.

4.5.1 Type 1

Integrals of the form $\int_{-\infty}^{\infty} \frac{P(x)}{Q(x)}\mathrm{d}x$, where $P(x)$ and $Q(x)$ are polynomials in x. This integral converges (exists), if

(i) The degree of $Q(x)$ is at least 2 greater than the degree of $P(x)$.

(ii) $Q(x)$ does not vanish for any real value of x.

To evaluate this integral, we evaluate $\int_C \frac{P(z)}{Q(z)}\mathrm{d}z$,

where C is the closed contour consisting of the real axis from $-R$ to $+R$ and the semicircle S above the real axis having this line as diameter, by using Cauchy's residue theorem and then letting $R \to \infty$. The contour C is shown in Fig. 4.22.

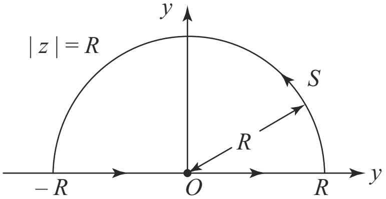

Fig. 4.22

Now we consider a result, known as Cauchy's Lemma, which will be used in evaluation of $\int_S f(z)\mathrm{d}z$, where S is the semicircle $|z| = R$ above the real axis.

Cauchy's Lemma

If $f(z)$ is a continuous function such that

$|zf(z)| \to 0$ uniformly as $|z| \to \infty$ on S, then $\int_S f(z)\mathrm{d}z \to 0,$ as $R \to \infty$, where S is the semicircle $|z| = R$ above the real axis.

Proof

$$\left|\int_S f(z)\mathrm{d}z\right| = \left|\int_S zf(z)\cdot\frac{1}{z}\mathrm{d}z\right|$$

$$\leq \int_S |zf(z)|\cdot\frac{1}{|z|}\cdot|\mathrm{d}z|$$

i. e. $$\int_0^{\pi} |zf(z)|\cdot\frac{1}{R}\cdot R\mathrm{d}\theta$$

[$\because$ When $|z| = R$, $z = Re^{i\theta}$ and $\mathrm{d}z = Rie^{i\theta}\mathrm{d}\theta$]

$\therefore$ $$\lim_{R\to\infty}\left|\int_S f(z)\mathrm{d}z\right| \leq \pi\times\lim_{R\to\infty}|zf(z)|$$

i. e. ≤ 0, by the given condition.

$\therefore$ $$\left|\int_S f(z)\mathrm{d}z\right| = 0 \text{ and hence } \int_S f(z)\mathrm{d}z = 0, \text{as } R\to\infty.$$

Note ☑ Similarly, we can shown that $\int_{S_1} f(z)\mathrm{d}z \to 0$ as $r\to 0$, where S_1 is the semicircle $|z - a| = r$ above the real axis, provided that $f(z)$ is a continuous function such that $|(z - a)f(z)| \to 0$ as $r \to 0$ uniformly.

4.5.2 Type 2

Integrals of the form $\int_{-\infty}^{\infty}\frac{P(x)\sin mx}{Q(x)}\mathrm{d}x$ or $\int_{-\infty}^{\infty}\frac{P(x)\cos mx}{Q(x)}\mathrm{d}x$, where $P(x)$ and $Q(x)$ are polynomials in x.

This integral converges (exists), if

(i) $m > 0$
(ii) the degree of $Q(x)$ is greater than the degree of $P(x)$
(iii) $Q(x)$ does not vanish for any real value of x. To evaluate this integral, we evaluate $\int_C \frac{P(z)}{Q(z)}e^{imz}\mathrm{d}z$, where C is the same contour as in Type 1, by using Cauchy's residue theorem and letting $R\to\infty$. After getting $\int_{-\infty}^{\infty}\frac{P(x)}{Q(x)}e^{imx}\mathrm{d}x$, we find the value of the real part or the imaginary part of this integral as per the requirement.

A result, known as Jordan's Lemma, which will be used in the evaluation of $\int_S f(z)e^{imz}\,dz$, where S is the semicircle $|z| = R$ above the real axis, is given as follows.

Jordan's Lemma

If $f(z)$ is a continuous function such that $|f(z)| \to 0$ uniformly as $|z| \to \infty$ on S, then $\int_S e^{imz} f(z)\,dz \to 0$ as $R \to \infty$, where S is he semicircle $|z| = R$ above the real axis and $m > 0$.

Proof

On the circle $|z|\ R$, $z = Re^{i\theta}$.

$$\therefore \quad \left|\int_S e^{imz} f(z)\,dz\right| = \left|\int_S e^{imR(\cos\theta + i\sin\theta)}.f(z).iRe^{i\theta}\,d\theta\right|$$

$$\leq \int_0^{\pi} \left|e^{imR(\cos\theta+i\sin\theta)}\right| |f(z)|\, R\,d\theta$$

i.e.
$$\leq \int_0^{\pi} e^{-mR\sin\theta}\,|f(z)|\,R\,d\theta$$

i.e.
$$\leq 2\int_0^{\pi/2} e^{-mR\sin\theta}\,|f(z)|\,R\,d\theta \tag{1}$$

Now
$$\sin\theta \geq \frac{2\theta}{\pi}, \text{for } 0 \leq \theta \leq \pi/2$$

$$\therefore \quad -mR\sin\theta \leq -\frac{2mR\theta}{\pi}, \text{for } 0 \leq \theta \leq \pi/2$$

$$\therefore \quad e^{-mR\sin\theta} \leq e^{-2mR\theta/\pi}, \text{ for } 0 \leq \theta \leq \pi/2 \tag{2}$$

Using (2) in (1), we have

$$\left|\int_S e^{imz} f(z)\,dz\right| \leq 2\int_0^{\pi/2} e^{-2mR\theta/\pi}\,|f(z)|\,R\,d\theta$$

$$\therefore \quad \lim_{R\to\infty}\left|\int_S e^{imz} f(z)\,dz\right| \leq \int_0^{\pi/2} \lim_{R\to\infty}(2Re^{-2mR\theta/\pi})\lim_{R\to\infty}[f(z)]\,d\theta$$

i.e.
$$\leq \lim_{R\to\infty}[f(z)] \times \lim_{R\to\infty}\int_0^{\pi/2} 2Re^{-2mR\theta/\pi}\,d\theta$$

i.e.
$$\leq \lim_{R\to\infty}[f(z)] \times \lim_{R\to\infty}\left[\frac{\pi}{m}\left(1-e^{-mR}\right)\right]$$

i.e. $\leq \dfrac{\pi}{m} \times 0$, by the given condition.

$$\therefore \quad \left| \int_S e^{imz} f(z) \mathrm{d}z \right| = 0 \text{ and hence } \int_S e^{imz} f(z) \mathrm{d}z = 0$$

as $R \to \infty$.

4.5.3 Type 3

Integrals of the form $\displaystyle\int_0^{2\pi} \frac{P(\sin\theta, \cos\theta)}{Q(\sin\theta, \cos\theta)} \mathrm{d}\theta$, where P and Q are polynomials in $\sin\theta$ and $\cos\theta$.

To evaluate this integral, we take the unit circle $|z| = 1$ as the contour.

When $|z| = 1$, $z = e^{i\theta}$ and so $\sin\theta = \dfrac{z - z^{-1}}{2i}$ and $\cos\theta = \dfrac{z + z^{-1}}{2}$. Also $\mathrm{d}z = e^{i\theta} i\mathrm{d}\theta$ or $\mathrm{d}\theta = \dfrac{\mathrm{d}z}{iz}$. When θ varies from 0 to 2π, the point z moves once around the unit circle $|z| = 1$.

Thus $\displaystyle\int_0^{2\pi} \frac{P(\sin\theta, \cos\theta)}{Q(\sin\theta, \cos\theta)} \mathrm{d}\theta = \int_C f(z)\, \mathrm{d}z,$

where $f(z)$ is a rational function of z and C is $|z| = 1$.

Now applying Cauchy's residue theorem, we can evaluate the integral on the R.H.S.

WORKED EXAMPLE 4(c)

Example 4.1 Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{x^2 \mathrm{d}x}{(x^2 + a^2)(x^2 + b^2)}$, using contour integration, where $a > b > 0$.

Consider $\displaystyle\int_C \frac{z^2 \mathrm{d}z}{(z^2 + a^2)(z^2 + b^2)}$, where C is the contour consisting of the segment of the real axis from $-R$ to $+R$ and the semicircle S above the real axis having this segment as diameter. The singularities of the integrand are given by

$$(z^2 + a^2)(z^2 + b^2) = 0$$

i.e. $z = \pm ia$ and $z = \pm ib$, which are simple poles.

Of these poles, only $z = ia$ and $z = ib$ lie inside C. (Fig. 4.23)

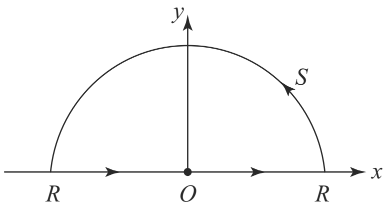

Fig. 4.23

Now $$R_1 = (\text{Res. of the integrand})_{z=ia} = \left[\frac{z^2}{(z+ia)(z^2+b^2)}\right]_{z=ia}$$

$$= \frac{-a^2}{2ia(b^2-a^2)} = \frac{a}{2i(a^2-b^2)}$$

Similarly $$R_2 = \text{Res. at } (z = ib) = -\frac{b}{2i(a^2-b^2)}$$

By Cauchy's residue theorem,

$$\int_C \frac{z^2}{(z^2+a^2)(z^2+b^2)}\,dz = 2\pi i(R_1+R_2)$$

$$= \frac{\pi(a-b)}{a^2-b^2} = \frac{\pi}{a+b}$$

i.e., $$\int_{-R}^{R} \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)} + \int_S \frac{z^2 dz}{(z^2+a^2)(z^2+b^2)} = \frac{\pi}{a+b} \qquad (1)$$

($\because$ on the real axis, $z = x$ and so $dz = dx$)

Now $|z^2 + a^2| \geq |z|^2 - a^2 = R^2 - a^2$ on $|z| = R$.

Similarly $|z^2 + b^2| \geq R^2 - b^2$ on $|z| = R$.

$$\therefore \quad \left|z.\frac{z^2}{(z^2+a^2)(z^2+b^2)}\right| \leq \frac{R^3}{(R^2-a^2)(R^2-b^2)}$$

Since the limit of the R.H.S. is zero as $R \to \infty$,

$$\lim_{R\to\infty}\left|z.\frac{z^2}{(z^2+a^2)(z^2+b^2)}\right| = 0, \text{on } |z| = R.$$

$\therefore$ By Cauchy's lemma, $$\int_S \frac{z^2 dz}{(z^2+a^2)(z^2+b^2)} = 0 \qquad (2)$$

Using (2) in (1) and letting $R \to \infty$, we get

$$\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)} = \frac{\pi}{a+b}$$

Example 4.2 Evaluate $\int\limits_{-\infty}^{\infty} \dfrac{x^2 dx}{(x^2+1^2)(x^2+2x+2)}$, using contour integration.

Consider $\int\limits_C \dfrac{z^2 dz}{(z^2+1^2)(z^2+2z+2)}$, where C is the same contour as in the previous example. The singularities of $f(z) = \dfrac{z^2}{(z^2+1)^2(z^2+2z+2)}$ are given by $(z^2 + 1)^2$ $(z^2 + 2z + 2) = 0$

i.e. $z = \pm i$ and $z = -1 \pm i$, of which $z = i$ and $z = -1 + i$ lie inside C.
$z = i$ is double pole and $z = -1 + i$ is a simple pole.

$$R_1 = [\text{Res.} f(z)_{z=i} = \frac{1}{1!}\left[\frac{d}{dz}\left\{\frac{(z-i)^2 \cdot z^2}{(z+i)^2(z-i)^2(z^2+2z+2)}\right\}\right]_{z=i}$$

$$= \left[\frac{(z+i)^2(z^2+2z+2).2z - z^2\{2(z+i)(z^2+2z+2)+(z+i)^2(2z+2)\}}{(z+i)^4(z^2+2z+2)^2}\right]_{z=i}$$

$$= \frac{9i-12}{100}$$

$$R_2 = [\text{Res.} f(z)]_{z=-1+i} = \left[\frac{z^2}{(z^2+1)^2(z+1+i)}\right]_{z=-1+i} = \frac{3-4i}{25}$$

By Cauchy's residue theorem,

$$\int\limits_C \frac{z^2 dz}{(z^2+1)^2(z^2+2z+2)} = 2\pi i(R_1 + R_2)$$

$$= 2\pi i\left\{\frac{9i-12}{100} + \frac{3-4i}{25}\right\}$$

$$= \frac{7\pi}{50}$$

i.e.
$$\int\limits_{-R}^{R} \frac{x^2 dx}{(x^2+1)^2(x^2+2x+2)} + \int\limits_S \frac{z^2 dz}{(z^2+1)^2(z^2+2z+2)} = \frac{7\pi}{50} \qquad (1)$$

$$\lim_{R\to\infty}\left|\frac{z.z^2}{(z^2+1)^2(z^2+2z+2)}\right| = 0 \text{ on } |z| = R.$$

$\therefore$ By Cauchy's Lemma, when $R \to \infty$

$$\int_S \frac{z^2 \mathrm{d}z}{(z^2+1)^2 (z+2z+2)} = 0 \tag{2}$$

Letting $R \to \infty$ in (1) and using (2), we get

$$\int_{-\infty}^{\infty} \frac{x^2 \mathrm{d}\,x}{(x^2+1)^2 (x^2+2x+2)} = \frac{7\pi}{50}$$

Example 4.3 Evaluate $\displaystyle\int_0^{\infty} \frac{\mathrm{d}x}{(x^2+a^2)^3}$, using contour integration, where $a > 0$.

Consider $\displaystyle\int_C \frac{\mathrm{d}z}{(z^2+a^2)^3}$, where C is the same contour as in example 4.1.

The singularities of $f(z) = \dfrac{1}{(z^2+a^2)^3}$ are $z = \pm\, ia$,which are poles of order 3. Of the two poles, $z = ia$ alone lies inside C.

$$R = [\mathrm{Res}.f(z)]_{z=ia} = \frac{1}{2!}\left[\frac{\mathrm{d}^2}{\mathrm{d}z^2}\left\{\frac{1}{(z+ia)^3}\right\}\right]_{z=ia}$$

$$= \left[\frac{6}{(z+ia)^5}\right]_{z=ia} = \frac{3}{16a^5 i}$$

$\therefore$ By Cauchy's residue theorem,

$$\int_C \frac{\mathrm{d}z}{(z^2+a^2)^3} = 2\pi i R = 2\pi i \,.\, \frac{3}{16a^5 i} = \frac{3\pi}{8a^5}$$

i.e.
$$\int_{-R}^{R} \frac{\mathrm{d}x}{(x^2+a^2)^3} + \int_S \frac{\mathrm{d}z}{(z^2+a^2)^3} = \frac{3\pi}{8a^5} \tag{1}$$

Now
$$\left| z.\frac{1}{(z^2+a^2)^3} \right| \le \frac{R}{(R^2-a^2)^3} \text{ and so}$$

$$\lim_{R\to\infty} \left| z.\frac{1}{(z^2+a^2)^3} \right| = 0 \text{ on } |z| = R$$

$\therefore$ By Cauchy's Lemma, when $R \to \infty$,

$$\int_S \frac{\mathrm{d}z}{(z^2+a^2)^3} = 0 \tag{2}$$

Letting $R \to \infty$ in (1) and using (2), we get

$$\int_{-\infty}^{\infty} \frac{dx}{(x^2+a^2)^3} = \frac{3\pi}{8a^5}.$$

Since the integrand $\frac{1}{(x^2+a^2)^3}$ is an even function of x, we have $\int_0^{\infty} \frac{dx}{(x^2+a^2)^3} = \frac{3\pi}{16a^5}$

Example 4.4 Use contour integration to prove that

$$\int_0^{\infty} \frac{x^2 dx}{x^4+a^4} = \frac{\pi}{2\sqrt{2}a}, \text{ where } a > 0.$$

Consider $\int_C \frac{z^2 dz}{z^4+a^4}$, where C is the same contour as in Example 4.1.

The singularities of $f(z) = \frac{z^2}{z^4+a^4}$ are given by $z^4 = -a^4 = e^{i(2r+1)\pi}.\ a^4$

i.e. $z = e^{i(2r+1)\pi/4}.\ a$, where $r = 0, 1, 2, 3.$

i.e. $z = ae^{i\pi/4}, ae^{i3\pi|4}, ae^{i5\pi/4}, ae^{i7\pi/4}$, all of which are simple poles.

Of these poles, only $z = ae^{i\pi/4}$ and $z = ae^{i3\pi/4}$ lie inside C.

$$R_1 = [\text{Res.} f(z)]_{z=aei\pi/4} = \lim_{z \to ae^{i\pi/4}} \left[\frac{z^2}{4z^3}\right] \left[= \lim_{z \to a} \frac{P(z)}{Q^1(z)}\right]$$

$$= \frac{1}{4a} e^{-i\pi/4} = \frac{1}{4\sqrt{2}a}(1-i)$$

$$R_2 = [\text{Res.} f(z)_{z=aei3\pi/4} = \frac{1}{4a} e^{-i3\pi/4} = \frac{1}{4\sqrt{2}a}(-1-i)$$

$\therefore$ By Cauchy's residue theorem,

$$\int_C \frac{z^2 dz}{z^4+a^4} = 2\pi i \cdot \frac{1}{4\sqrt{2}a}(1-i-1-i) = \frac{\pi}{\sqrt{2}a}$$

i.e. $$\int_{-R}^{R} \frac{x^2 dx}{x^4+a^4} + \int_S \frac{z^2 dz}{z^4+a^4} = \frac{\pi}{\sqrt{2}a} \tag{1}$$

Now $$\lim_{r \to \infty} \left| z.\frac{z^2}{z^4+a^4} \right| = 0 \text{ on } |z| = R.$$

$\therefore$ By Cauchy's Lemma, when $R \to \infty$,

$$\int_s \frac{z^2 dz}{z^4+a^4} = 0 \tag{2}$$

Letting $R \to \infty$ in (1) and using (2), we get

$$\int_{-\infty}^{\infty} \frac{x^2 dx}{x^4 + a^4} = \frac{\pi}{\sqrt{2}a}$$

Since the integrand is even,

$$\int_{0}^{\infty} \frac{x^2 dx}{x^4 + a^4} = \frac{\pi}{2\sqrt{2}a}$$

Example 4.5 Evaluate $\int_{-\infty}^{\infty} \frac{x^4}{x^6 - a^6} dx$, using contour integration where $a > 0$.

Consider $\int_C \frac{z^4}{z^6 - a^6} dz$, where C is the same contour as in example 4.1, with some modifications explained below.

The singularities of $f(z) = \frac{z^4}{z^6 - a^6}$ are given by

$$z^6 = a^6 = e^{i2r\pi} a^6, \text{ i.e. } z = ae^{\frac{r\pi}{3}i}, r = 0, 1, 2, 3, 4, 5.$$

i.e. $z = a, ae^{\pi i/3}, ae^{2\pi i/3}, -a, ae^{4\pi i/3}, ae^{5\pi i/3}$.

Of these singularities (simple poles),

$z = ae^{\pi i/3}$ and $z = ae^{2\pi i/3}$ lie inside C, but $z = a$ and $z = -a$ lie on the real axis. But for the evaluation of the integrals of type 1, no singularity of $f(z)$ should lie on the real axis. To avoid them, we modify C by introducing small indents, i.e., semi-circle of small radius at $z = \pm a$, as shown in Fig. 4.24.

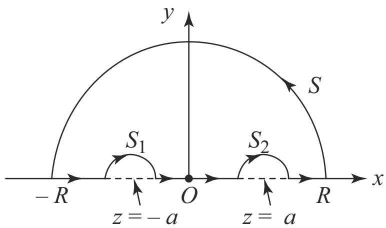

Fig. 4.24

Now the modified contour C contains only the simple poles

$$z = ae^{\pi i/3} \quad \text{and} \quad ae^{2\pi i/3}$$

$$R_1 = [\text{Res. } f(z)]_{z=ae^{\pi/3i}}$$

$$= \left(\frac{z^4}{6z^5}\right) = \frac{1}{6a} e^{-\pi i/3}$$

$$= \frac{1}{6a}\left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right)$$

Similarly, $$R_2 = \frac{1}{6a} e^{-2\pi i/3} = \frac{1}{6a}\left(-1/2 - i\frac{\sqrt{3}}{2}\right)$$

By Cauchy's residue theorem,

$$\int_C \frac{z^4}{z^6 - a^6} \mathrm{d}z = \frac{2\pi i}{6a}\left(1/2 - i\frac{\sqrt{3}}{2} - 1/2 - i\frac{\sqrt{3}}{2}\right)$$

$$= \frac{\pi}{\sqrt{3}a}$$

i.e. $$\int_{-R}^{-a-r} \frac{x^4}{x^6 - a^6} \mathrm{d}x + \oint_{S_1} \frac{z^4}{z^6 - a^6} \mathrm{d}z + \int_{-a+r}^{a-r'} \frac{x^4}{x^6 - a^6} \mathrm{d}x + \oint_{S_2} \frac{z^4}{z^6 - a^6} \mathrm{d}z$$

$$+ \int_{a+r'}^{R} \frac{x^4}{x^6 - a^6} \mathrm{d}x + \oint_S \frac{z^4}{z^6 - a^6} \mathrm{d}z = \frac{\pi}{\sqrt{3}a} \tag{1}$$

where r and r' are the radii of the semicircles S_1 and S_2 whose equations are $|z + a| = r$ and $|z - a| = r'$. These two integrals taken along S_1 and S_2 vanish as $r \to 0$ and $r' \to 0$, by the note under Cauchy's Lemma.

$$\int_S \frac{z^4}{z^6 - a^6} = 0 \text{, as } R \to \infty \text{, by Cauchy's lemma.}$$

Now, letting $r \to 0$, $r' \to 0$ and $R \to \infty$ in (1),

We get $$\int_{-\infty}^{-a} + \int_{-a}^{a} + \int_{a}^{\infty} \frac{x^4 \mathrm{d}x}{x^6 - a^6} = \frac{\pi}{\sqrt{3}a}$$

i.e. $$\int_{-\infty}^{\infty} \frac{x^4}{x^6 - a^6} \mathrm{d}x = \frac{\pi}{\sqrt{3}a}$$

Example 4.6 Evaluate $\int_0^\infty \frac{x \sin x}{x^2 + a^2} \mathrm{d}x$, by contour integration.

Consider $\int_C \frac{z e^{iz}}{z^2 + a^2} \mathrm{d}z$, where C is the same contour as in Example 4.1.

The singularities of $f(z) = \frac{z e^{iz}}{z^2 + a^2}$. are $z = \pm\, ia$, which are simple poles. Of these poles, only $z = ia$ lies inside C.

$$[\text{Res.} f(z)]_{z = ia} = \left(\frac{z e^{iz}}{z + ia}\right)_{z = ia} = \frac{1}{2} e^{-a}.$$

By Cauchy's residue theorem,

$$\int_C \frac{ze^{iz}}{z^2+a^2}\,\mathrm{d}z = 2\pi i\,.\,\frac{1}{2}e^{-a} = \pi i e^{-a}$$

i.e. $$\int_{-R}^{R} \frac{xe^{ix}}{x^2+a^2}\,\mathrm{d}x + \int_S \frac{ze^{iz}}{z+a^2}\,\mathrm{d}z = \pi i e^{-a} \qquad (1)$$

Now $$\left|\frac{z}{z^2+a^2}\right| \le \frac{R}{R^2-a^2}$$

Since the limit of the R.H.S. is zero as $R \to \infty$,

$$\lim_{R\to\infty}\left|\frac{z}{z^2+a^2}\right| = 0 \text{ on } |z| = R.$$

$\therefore$ By Jordan's Lemma, $\displaystyle\int_S \frac{ze^{iz}}{z^2+a^2}\,\mathrm{d}z \;\to 0$ as $R \to \infty$, (2)

Letting $R\to\infty$ in (1) and using (2), we get

$$\int_{-\infty}^{\infty} \frac{xe^{ix}}{x^2+a^2}\,\mathrm{d}x = i\pi e^{-a}$$

i.e. $$\int_{-\infty}^{\infty} \frac{x}{x^2+a^2}(\cos x + i\sin x)\mathrm{d}x = i\pi e^{-a}.$$

Equating the imaginary parts on both sides,

we get $$\int_{-\infty}^{\infty} \frac{x\sin x}{x^2+a^2}\,\mathrm{d}x = \pi e^{-a}.$$

Since the integrand is an even function of x,

$$\int_0^{\infty} \frac{x\sin x}{x^2+a^2}\,\mathrm{d}x = \frac{\pi}{2}e^{-a}.$$

Example 4.7 Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{\cos x\mathrm{d}x}{\left(x^2+a^2\right)\left(x^2+b^2\right)}$., using contour integration, where a > b > 0.

Consider $\displaystyle\int_C \frac{e^{iz}}{\left(z^2+a^2\right)\left(z^2+b^2\right)}\mathrm{d}z$, where C is the same contour as in Example 4.1.

The singularities of the integrand $f(z)$ are given by $z = \pm\, ia$ and $z = \pm\, ib$, which are simple poles. Of these poles, $z = ia$ and $z = ib$ only lie inside C.

$$R_1 = [\text{Res.} f(z)]_{z=ia} = \left[\frac{e^{iz}}{(z+ia)\left(z^2+b^2\right)}\right]_{z=ia}$$

$$= \frac{e^{-a}}{-2ia\left(a^2-b^2\right)}$$

$$R_2 = [\text{Res.} f(z)]_{z=ib} = \frac{e^{-b}}{2ib(a^2-b^2)}$$

By Cauchy's residue theorem,

$$\int_C \frac{e^{iz}}{\left(z^2+a^2\right)\left(z^2+b^2\right)} \mathrm{d}z = 2\pi i\left(R_1+R_2\right)$$

$$= \frac{\pi}{\left(a^2-b^2\right)}\left(\frac{e^{-b}}{b} - \frac{e^{-a}}{a}\right)$$

i.e.
$$\int_{-R}^{R} \frac{e^{ix}}{\left(x^2+a^2\right)\left(x^2+b^2\right)} \mathrm{d}x + \int_S \frac{e^{iz}}{\left(z^2+a^2\right)\left(z^2+b^2\right)} \mathrm{d}z$$

$$= \frac{\pi}{a^2-b^2}\left(\frac{e^{-b}}{b} - \frac{e^{-a}}{a}\right) \tag{1}$$

Now
$$\left|\frac{1}{\left(z^2+a^2\right)\left(z^2+b^2\right)}\right| \le \frac{1}{(R^2-a^2)\left(R^2-b^2\right)}$$

R.H.S. $\to 0$ as $R \to \infty$. Hence

$$\lim_{R\to\infty}\left|\frac{1}{\left(z^2+a^2\right)\left(z^2+b^2\right)}\right| = 0 \text{ on } |z| = R.$$

$\therefore$ By Jordan's Lemma, $\displaystyle\int_S \frac{e^{iz}}{\left(z^2+a^2\right)\left(z^2+b^2\right)} \mathrm{d}z \to 0$, as $R \to \infty$ (2)

Letting $R\to\infty$ in (1) and using (2), we get

$$\int_{-\infty}^{\infty} \frac{e^{ix}}{\left(x^2+a^2\right)\left(x^2+b^2\right)} \mathrm{d}x = \frac{\pi}{\left(a^2-b^2\right)}\left(\frac{e^{-b}}{b} - \frac{e^{-a}}{a}\right)$$

Equating the real parts on both sides, we get

$$\int_{-\infty}^{\infty} \frac{\cos x}{\left(x^2+a^2\right)\left(x^2+b^2\right)} \mathrm{d}x = \frac{\pi}{\left(a^2-b^2\right)}\left(\frac{e^{-b}}{b} - \frac{e^{-a}}{a}\right).$$

Example 4.8 Evaluate $\int_0^\infty \frac{\cos ax}{(x^2+b^2)} \mathrm{d}x = \frac{\pi}{4b^3}(1+ab)e^{-ab}$, where $a > 0$ and $b > 0$.

Consider $\int_C \frac{e^{iaz}}{(z^2+b^2)^2} \mathrm{d}z$, where C is the same contour as in Example 4.1.

The singularities of $f(z) = \frac{e^{iaz}}{(z^2+b^2)^2}$ are $z = \pm\, ib$, which are double poles. Of these poles, only $z = ib$ lies inside C.

$$[\text{Res. } f(z)]_{z=ib} = \frac{1}{1!}\frac{\mathrm{d}}{\mathrm{d}z}\left[\left(\frac{e^{iaz}}{(z+ib)^2}\right)\right]_{z=ib}$$

$$= \left[\frac{(z+ib)ia\,e^{iaz} - 2e^{iaz}}{(z+ib)^3}\right]_{z=ib}$$

$$= \frac{1}{4ib^3}(ab+1)e^{-ab}.$$

By Cauchy's residue theorem,

$$\int_{-R}^{R} \frac{e^{iax}}{(x^2+b^2)^2}\mathrm{d}x + \int_S \frac{e^{iaz}}{(z^2+b^2)^2}\mathrm{d}z = 2\pi i \times \frac{1}{4ib^3}(ab+1)e^{-ab}$$

$$= \frac{\pi}{2b^3}(ab+1)e^{-ab} \tag{1}$$

Now $\left|\frac{1}{(z^2+b^2)^2}\right| \leq \frac{1}{(R^2-b^2)^2}$

Since the R.H.S.$\rightarrow 0$ as $R \rightarrow \infty$, L.H.S. also $\rightarrow 0$ as $R \rightarrow \infty$ on $|z| = R$.

$\therefore$ By Jordan's Lemma,

$$\int_S \frac{e^{iaz}}{(z^2+b^2)^2}\mathrm{d}z \rightarrow 0 \text{ as } R \rightarrow \infty, \text{ since } a > 0 \tag{2}$$

Letting $R \rightarrow \infty$ in (1) and using (2), we get

$$\int_{-\infty}^{\infty} \frac{e^{iax}}{(x^2+b^2)^2}\mathrm{d}x = \frac{\pi}{2b^3}(ab+1)e^{-ab}.$$

Equating the real parts on both sides and noting that $\dfrac{\cos ax}{(x^2+b^2)^2}$ is an even function of x, we get

$$\int_0^\infty \frac{\cos ax}{(x^2+b^2)^2}\,dx = \frac{\pi}{4b^3}(ab+1)e^{-ab}.$$

Example 4.9 Use contour integration to prove that

$$\int_0^\infty \frac{\sin mx}{x}\,dx = \frac{\pi}{2}, \text{ when } m > 0.$$

Consider $\int_C \dfrac{e^{imz}}{z}dz$, where C is the usual semicircular contour, but with an indent i.e. a small semicircle at the origin, which is introduced to avoid the singularity $z = 0$, which lies on the real axis. The modified contour is shown in Fig. 4.25.

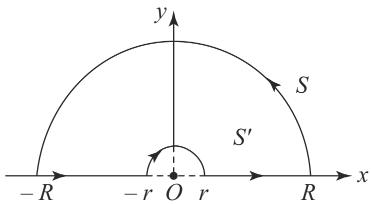

Fig. 4.25

The modified contour C does not include any singularity of $f(z) = \dfrac{e^{imz}}{z}$.
∴ By Cauchy's residue theorem,

$$\int_{-R}^{-r} \frac{e^{imx}}{x}\,dx + \oint_{S'} \frac{e^{imz}}{z}\,dz + \int_r^R \frac{e^{imx}}{x}\,dx + \oint_S \frac{e^{imz}}{z}\,dz = 0 \qquad (1)$$

The Eqn. of S' is $|z| = r$. ∴ $z = re^{i\theta}$ and $dz = re^{i\theta}\, i d\theta$.

As S' is described in the clockwise sense, θ varies from π to 0.

Thus $$\int_{S'} \frac{e^{imz}}{z}\,dz = \int_\pi^0 \frac{e^{imre^{i\theta}}}{re^{i\theta}} re^{i\theta} i d\theta$$

∴ $$\lim_{r\to 0}\int_{S'} \frac{e^{imz}}{z}\,dz = \int_\pi^0 \left[\lim_{r\to 0}\left(e^{imre^{i\theta}}\right)\right] i d\theta = -i\pi \qquad (2)$$

$$\left|\frac{1}{z}\right| = \frac{1}{R} \to 0 \text{ as } R\to\infty \text{ on } |z| = R.$$

∴ By Jordan's Lemma, $\int_S \dfrac{e^{imz}}{z}dz \to 0$ as $R \to \infty$, since $m > 0$ (3)

Letting $r \to 0$ and $R \to \infty$ in (1) and using (2) and (3),

we get $$\int_{-\infty}^{0} \frac{e^{imx}}{x}\,dx - i\pi + \int_{0}^{\infty} \frac{e^{imx}}{x}\,dx = 0$$

i.e. $$\int_{-\infty}^{\infty} \frac{e^{imx}}{x}\,dx = i\pi$$

Equating the imaginary parts on both sides and noting that $\dfrac{\sin mx}{x}$ is an even function of x, we get $\displaystyle\int_{0}^{\infty} \frac{\sin mx}{x}\,dx = \frac{\pi}{2}$.

Example 4.10 Evaluate $\displaystyle\int_{0}^{\infty} \frac{\sin x\,dx}{x\left(x^2+a^2\right)}$, using contour integrations where $a > 0$.

Consider $\displaystyle\int_{C} \frac{e^{iz}}{z\left(z^2+a^2\right)}\,dz$, where C is the same modified contour as in Example 4.9, which includes the only pole $z = ia$ of $f(z) = \dfrac{e^{iz}}{z\left(z^2+a^2\right)}$

$$[\text{Res.}\, f(z)]_{z=ia} = \left[\frac{e^{iz}}{z(z+ia)}\right]_{z=ia} = \frac{e^{-a}}{-2a^2}$$

By Cauchy's residue theorem,

$$\int_{-R}^{-r} \frac{e^{ix}}{x\left(x^2+a^2\right)}\,dx + \oint_{S'} \frac{e^{iz}}{z\left(z^2+a^2\right)}\,dz + \int_{r}^{R} \frac{e^{ix}}{x\left(x^2+a^2\right)}\,dx + \oint_{S} \frac{e^{iz}}{z\left(z^2+a^2\right)}\,dz$$

$$= -\frac{\pi i}{a^2} e^{-a} \tag{1}$$

$$\oint_{S'} \frac{e^{iz}}{z\left(z^2+a^2\right)}\,dz = \int_{\pi}^{0} \frac{e^{ir e^{i\theta}}.re^{i\theta}\,id\theta}{re^{i\theta}\left(r^2 e^{i2\theta}+a^2\right)}$$

$\therefore$ $$\lim_{r\to 0}\int_{S'} \frac{e^{iz}}{z\left(z^2+a^2\right)}\,dz = \int_{\pi}^{0} \lim_{r\to 0}\left[\frac{e^{ir e^{i\theta}}}{r^2 e^{i2\theta}+a^2}\right] id\theta$$

$$= -\frac{\pi i}{a^2} \tag{2}$$

$$\left|\frac{1}{z\left(z^2+a^2\right)}\right| \le \frac{1}{R\left(R^2-a^2\right)}$$

Since R.H.S. $\to 0$ as $R \to \infty$ on $|z| = R$, L.H.S. also tends to 0 as $R \to \infty$.

$\therefore$ By Jordan's Lemma,

$$\oint_S \frac{e^{iz}}{z\left(z^2+a^2\right)}\,\mathrm{d}z \to 0, \text{ as } R \to \infty \tag{3}$$

Letting $r \to 0$ and $R \to \infty$ in (1) and using (2) and (3), we get

$$\int_{-\infty}^{\infty} \frac{e^{ix}}{x\left(x^2+a^2\right)}\,\mathrm{d}x = \frac{\pi i}{a^2} - \frac{\pi i}{a^2} e^{-a}.$$

Equating imaginary parts on both sides and noting that $\dfrac{\sin x}{x\left(x^2+a^2\right)}$ is an even function of x, we get,

$$\int_0^{\infty} \frac{\sin x}{x\left(x^2+a^2\right)}\,\mathrm{d}x = \frac{\pi}{2a^2}\left(1-e^{-a}\right).$$

Example 4.11 Evaluate $\displaystyle\int_0^{2\pi} \frac{\mathrm{d}\theta}{1-2x\sin\theta+x^2}\ (0<x<1)$, using contour integration.

On the circle $|z| = 1$, $z = e^{i\theta}$, $\mathrm{d}z = ie^{i\theta}\,\mathrm{d}\theta$ or $\mathrm{d}\theta = \dfrac{\mathrm{d}z}{iz}$ and $\sin\theta = \dfrac{z-\frac{1}{z}}{2i} = \dfrac{z^2-1}{2iz}$.

$\therefore$ The given integral $I = \displaystyle\int_C \frac{\mathrm{d}z/iz}{1-2x\left(\dfrac{z^2-1}{2iz}\right)+x^2}$, where C is $|z| = 1$

i.e. $$I = \int_C \frac{\mathrm{d}z}{iz - xz^2 + x + ix^2 z}$$

$$= -\frac{1}{x}\int_C \frac{\mathrm{d}z}{z^2 - i\left(x+\dfrac{1}{x}\right)z - 1} = -\frac{1}{x}\int_C \frac{\mathrm{d}z}{(z-ix)\left(z-\dfrac{i}{x}\right)} \tag{1}$$

The singularities of $\dfrac{1}{(z-ix)\left(z-\dfrac{i}{x}\right)}$ are $z = ix$ and $z = \dfrac{i}{x}$, which are simple poles

Now $|ix| = |x| < 1$, as $0 < x < 1$

$\therefore$ The pole $z = ix$ lies inside C, but $z = \dfrac{i}{x}$ lies outside C.

$$\left[\text{Res.}\left\{\frac{1}{(z-ix)(z-i/x)}\right\}\right]_{z=ix} = \frac{1}{i\left(x-\dfrac{1}{x}\right)} = \frac{ix}{1-x^2}$$

Using Cauchy's residue theorem, from (1) we get

$$I = -\frac{1}{x} \times 2\pi i \times \frac{ix}{1-x^2} = \frac{2\pi}{1-x^2}.$$

Note ☑ We have integrated w.r.t. θ and x is a parameter.

Example 4.12 Evaluate $\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta} (a > b > 0),$ using contour integration.

Deduce the value of $\int_0^{2\pi} \frac{d\theta}{(a+b\cos\theta)^2}$. On the circle $|z| = 1$, $z = e^{i\theta}$, $d\theta = \frac{dz}{iz}$ and $\cos\theta = \frac{z^2+1}{2z}$.

$$\therefore I = \int_0^{2\pi} \frac{d\theta}{a+b\cos\theta} = \int_C \frac{dz/iz}{a+b\frac{z^2+1}{2z}} = -\frac{2i}{b}\int_C \frac{dz}{\left(z^2+\frac{2a}{b}z+1\right)} \quad (1)$$

where C is $|z| = 1$.

The singularities of the integrand are given by $z^2 + \frac{2a}{b}z + 1 = 0,$

i.e. $z = -\frac{a}{b} \pm \sqrt{\frac{a^2}{b^2} - 1},$ which are simple poles.

Since $a > b, \frac{a}{b} > 1$ and hence $\left| -\frac{a}{b} - \sqrt{\frac{a^2}{b^2}-1} \right| > 1$ and so $z = -\frac{a}{b} - \sqrt{\frac{a^2}{b^2}-1}$ lies outside $|z| = 1$.

$z = -\frac{a}{b} + \sqrt{\frac{a^2}{b^2}-1}$ lies inside $|z| = 1$.

$$\left[\text{Res. of} \left(\frac{1}{z^2+\frac{2a}{b}z+1}\right)\right]_{z=-\frac{a}{b}+\sqrt{\frac{a^2}{b^2}-1}}$$

$$= \left(\frac{1}{2\left(z+\frac{a}{b}\right)}\right)_{z=-\frac{a}{b}+\sqrt{\frac{a^2}{b^2}-1}} \quad (2)$$

$$= \frac{b}{2\sqrt{a^2-b^2}}$$

By Cauchy's residue theorem and using (2) in (1), we get

$$I = -\frac{2i}{b} \times 2\pi i \frac{b}{2\sqrt{a^2-b^2}} = \frac{2\pi}{\sqrt{a^2-b^2}}$$

i.e. $$\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta} = \frac{2\pi}{\sqrt{a^2-b^2}} \qquad (3)$$

Differentiating both sides of (3) partially with respect to 'a', we get

$$\int_0^{2\pi} -\frac{d\theta}{(a+b\cos\theta)^2} = -\frac{2\pi a}{\left(a^2-b^2\right)^{3/2}}$$

$$\therefore \quad \int_0^{2\pi} \frac{d\theta}{(a+b\cos\theta)^2} = \frac{2\pi a}{\left(a^2-b^2\right)^{3/2}}$$

Example 4.13 Evaluate $\int_0^{2\pi} \frac{\sin^2\theta}{5-3\cos\theta}\, d\theta$, using contour integration.

On the circle $|z| = 1$, $z = e^{i\theta}$, $d\theta = \frac{dz}{iz}$, $\sin\theta = \frac{z^2-1}{2iz}$ and $\cos\theta = \frac{z^2+1}{2z}$.

$$\therefore \quad I = \int_0^{2\pi} \frac{\sin^2\theta}{5-3\cos\theta}\, d\theta = \int_C \frac{\frac{\left(z^2-1\right)^2}{-4z^2} \cdot \frac{dz}{iz}}{5 - \frac{3\left(z^2+1\right)}{2z}}, \text{ where } C \text{ is } |z| = 1$$

$$= -\frac{i}{6} \int_C \frac{\left(z^2-1\right)^2 dz}{z^2 (z-3)(z-1/3)} \qquad (1)$$

The singularities of the integrand which lie within C are $z = 0$ and $z = 1/3$.

$z = 0$ is a double pole and $z = \frac{1}{3}$ is a simple pole.

$$R_1 = [\text{Res. of the integrand } f(z)]_{z=0}$$

$$= \frac{1}{1!}\left[\frac{d}{dz}\left\{\frac{\left(z^2-1\right)^2}{z^2 - \frac{10}{3}z + 1}\right\}\right]_{z=0} = \frac{10}{3}$$

$$R_2 = [\text{Res. } f(z)]_{z=1/3} = \left[\frac{\left(z^2-1\right)^2}{z^2(z-3)}\right]_{z=1/3} = -\frac{8}{3}$$

By Cauchy's residue theorem, from (1), we get

$$I = -\frac{i}{6}\times 2\pi i\left(R_1 + R_2\right) = \frac{\pi}{3}\left(\frac{10}{3}-\frac{8}{3}\right) = \frac{2\pi}{9}.$$

Example 4.14 Evaluate $\int_0^{2\pi} \frac{\cos 2\theta \, d\theta}{1-2a\cos\theta + a^2}$, using contour integration, where $a^2 < 1$.

On the circle $|z| = 1$, $z = e^{i\theta}$, $d\theta = \frac{dz}{iz}$, $\cos\theta = \frac{z^2+1}{2z}$ and $\cos 2\theta = \frac{e^{i2\theta}+e^{-i2\theta}}{2} = \frac{z^4+1}{2z^2}$.

$\therefore$ The given integral $I = \int_C \frac{\frac{z^4+1}{2z^2}\frac{dz}{iz}}{1-a\left(\frac{z^2+1}{z}\right)+a^2}$, where C is $|z| = 1$

$$= \frac{i}{2a}\int_C \frac{\left(z^4+1\right)dz}{z^2\left\{z^2-\left(a+\frac{1}{a}\right)z+1\right\}}$$

$$= \frac{i}{2a}\int_C \frac{\left(z^4+1\right)dz}{z^2\,(z-a)\left(z-1/a\right)} \qquad (1)$$

The singularities of the integrand which lie inside C are $z = 0$ and $z = a$ ($\because a^2 < 1$)
$z = 0$ is a double pole and $z = a$ is a simple pole.
R_1 = [Res. of the integrand]$_{z=0}$

$$= \frac{1}{1!}\left[\frac{d}{dz}\left\{\frac{z^4+1}{z^2-(a+1/a)z+1}\right\}\right]_{z=0}$$

$$= a + \frac{1}{a}$$

$$R_2 = (\text{Res. of the integrand})_{z=a} = \frac{a^4+1}{a^2\,(a-1/a)}$$

$$\therefore \qquad R_1 + R_2 = a + \frac{1}{a} + \frac{a^4+1}{a^2\,(a-1/a)}$$

$$= \frac{2a^3}{a^2-1}$$

By Cauchy's residue theorem and from (1), we get

$$I = \frac{i}{2a}\times 2\pi i \times \frac{2a^3}{a^2-1} = \frac{2\pi a^2}{1-a^2}$$

Example 4.15 Evaluate $\int_0^{2\pi} \frac{\cos 3\theta}{5+4\cos\theta} d\theta$, using contour integration

On the circle $|z| = 1$, $z = e^{i\theta}$, $d\theta = \frac{dz}{iz}$, $\cos\theta = \frac{z^2+1}{2z}$ and $\cos 3\theta = \frac{e^{i3\theta}+e^{-i3\theta}}{2} = \frac{z^6+1}{2z^3}$.

$\therefore$ The given integral $I = \int_C \frac{\frac{z^6+1}{2z^3} \cdot \frac{dz}{iz}}{5+4\left(\frac{z^2+1}{2z}\right)}$, where C is $|z| = 1$.

i.e.
$$I = -\frac{i}{4} \int_C \frac{(z^6+1)\,dz}{z^3(z+2)(z+1/2)}$$

The singularities of the integrand $f(z)$ which lie inside C are the simple pole $z = -1/2$ and the triple pole $z = 0$.

$R_1 = (\text{Res. of the integrand})_{z=-1/2} = \left\{\frac{z^6+1}{z^3(z+2)}\right\}_{z=-1/2} = -\frac{65}{12}$

R_2, the residue at $z = 0$ is found out as the coefficient of $\frac{1}{z}$ in the Laurent's expansion of the integrand.

$$f(z) = \left(z^3 + \frac{1}{z^3}\right)\left(1+\frac{z}{2}\right)(1+2z)$$

$$= \left(z^3 + \frac{1}{z^3}\right)\left\{1-\frac{z}{2}+\frac{z^2}{4}-\cdots\right\}\{1-2z+4z^2-\cdots\}$$

$$= \left(z^3 + \frac{1}{z^3}\right)\left\{1-\frac{5}{2}z+\frac{21}{4}z^2-\cdots\right\}$$

$\therefore R_2 = \text{Coefficient of } \frac{1}{z} \text{ in this expansion} = \frac{21}{4}$.

By Cauchy's residue theorem.

$$I = -i/4 \times 2\pi i\left(-\frac{65}{12}+\frac{21}{4}\right) = -\frac{\pi}{12}$$

EXERCISE 4(c)

Part A

(Short Answer Questions)

1. Give the forms of the definite integrals which can be evaluated using the infinite semi-circular contour above the real axis.
2. Explain how to convert $\int_0^{2\pi} f(\sin\theta, \cos\theta)\mathrm{d}\theta$ into a contour integral, where f is a rational function.
3. Sketch the contour to be used for the evaluation of $\int_0^{\infty} \frac{\sin mx}{x}\mathrm{d}x$.

Part B

Evaluate the following integrals by contour integration technique.

4. $\int_0^{\infty} \frac{x^2\mathrm{d}x}{(x^2+1)(x^2+4)}$
5. $\int_0^{\infty} \frac{x^2\mathrm{d}x}{(x^2+4)^2(x^2+9)}$
6. $\int_{-\infty}^{\infty} \frac{x^2-x+2}{x^4+10x^2+9}\mathrm{d}x$
7. $\int_0^{\infty} \frac{\mathrm{d}x}{x^4+a^4}\mathrm{d}x$
8. $\int_0^{\infty} \frac{x^2\mathrm{d}x}{(x^2+1)^3}$
9. $\int_0^{\infty} \frac{x^2}{x^6+1}\mathrm{d}x$
10. $\int_0^{\infty} \frac{x^4\mathrm{d}x}{x^6+1}$
11. $\int_{-\infty}^{\infty} \frac{\cos ax}{x^2+b^2}\mathrm{d}x\,(a>0; b>0)$
12. $\int_{-\infty}^{\infty} \frac{\cos x\,\mathrm{d}x}{(x^2+4)(x^2+9)}$
13. $\int_0^{\infty} \frac{x\sin x}{x^4+1}\mathrm{d}x$
14. $\int_{-\infty}^{\infty} \frac{x\sin x}{(x^2+1)(x^2+4)}\mathrm{d}x$
15. $\int_{-\infty}^{\infty} \frac{x\sin x}{(x^2+1)^2}\mathrm{d}x$
16. $\int_{-\infty}^{\infty} \frac{\sin x\,\mathrm{d}x}{x^2+4x+5}$
17. $\int_{-\infty}^{\infty} \frac{x\cos \pi x}{x^2+2x+5}\mathrm{d}x$
18. $\int_0^{\infty} \frac{\sin x}{x(x^2+1)^2}\mathrm{d}x$
19. $\int_0^{2\pi} \frac{\mathrm{d}\theta}{1-2p\cos\theta+p^2}, 0<p<1$

20. $\int_0^{2\pi} \frac{d\theta}{a+b\sin\theta}(a>b>0)$

21. $\int_0^{2\pi} \frac{\sin^2\theta}{a+b\cos\theta} d\theta (a>b>0)$

22. $\int_0^{2\pi} \frac{a\,d\theta}{a^2+\sin^2\theta}(a>0)$

23. $\int_0^{2\pi} \frac{\cos 2\theta\, d\theta}{5+4\cos\theta}$

24. $\int_0^{2\pi} \frac{\cos 3\theta\, d\theta}{5-4\cos\theta}$

25. $\int_0^{2\pi} \frac{\cos^2 3\theta}{5-4\cos 2\theta} d\theta$

ANSWERS

Exercise 4(a)

8. $\frac{10}{3}(3+i)$

9. $3 - i2$

10. $4\pi i$

11. $-\frac{4}{3}+\frac{8}{3}i$

12. πi

13. (i) $-1/6 + i5/6$; (ii) $-1/6+i\frac{13}{15}$.

14. $4\pi i$

15. $10-i\frac{8}{3}$

16. 0 in all cases.

17. (i) $\frac{511}{3}-\frac{49}{5}i$; (ii) $\frac{518}{3}-57i$; (iii) $\frac{518}{3}-8i$.

18. $\frac{1}{15}\left(96\pi^5a^5+80\pi^3a^3+30\pi a\right)$

19. (i) 0; (ii) $2\pi i$.

20. $\frac{8\pi i}{3}$

21. $-4\pi i$

22. (i) $2\pi i$; (ii) 0

23. $-\frac{2\pi}{3}$

24. $\sin t$

25. $\frac{8\pi i}{3e^2}$

26. $\frac{\pi}{16}$

27. $\frac{\pi}{2}(3+2i)$

28. 0

29. $20\pi i$; $2\pi(i-1)$; $-14\pi i$; $16\pi i$

Exercise 4(b)

17. $\sin z = \frac{1}{\sqrt{2}}\left\{1+(z-\pi/4)-\frac{(z-\pi/4)^2}{2!}-\frac{(z-\pi/4)^3}{3!}+\cdots\right\}$

18. $\cos z = \frac{1}{2}\left[1-\sqrt{3}\left(z-\pi/3\right)-\frac{1}{2!}\left(z-\pi/3\right)^2+\frac{\sqrt{3}}{3!}\left(z-\pi/3\right)^3 + \frac{1}{4!}\left(z-\pi/3\right)^4-\cdots\right]$

19. $e^{-i}\left\{1+\frac{(z+i)}{1!}+\frac{(z+i)^2}{2!}+\cdots\right\}$

20. $e^{-1}\left\{1-\frac{(z-1)}{1!}+\frac{(z-1)^2}{2!}-\frac{(z-1)^3}{3!}+\cdots\right\}$

21. $(z-1)-2(z-1)^2+3(z-1)^3-4(z-1)^4+\cdots$

22. $|z| < 1.$

23. $|z| < 2.$

24. $|z-1| < \frac{1}{2}.$

25. $|z-2| < 1$

26. $|z-4i| < 4.$

27. $-\frac{1}{2z^2}\left(1+\frac{z}{2}+\frac{z^2}{4}+\cdots\right)$

28. $\frac{1}{z-1}\left\{1-(z-1)+(z-1)^2-(z-1)^3+\cdots\right\}$

29. $-\frac{1}{z^4}\left(1+\frac{1}{z}+\frac{1}{z^2}+\cdots\right)$

30. $\frac{1}{z-1}\left[1-\frac{2}{z-1}+\frac{2}{(z-1)^2}-\cdots\right]$

31. 1

32. –1

33. 1

34. $\frac{1}{6}.$

35. $-\frac{1}{2}$

36. $R_{(z=-1)}=\frac{1}{3}; R_{(z=2)}=\frac{2}{3}.$

37. $\pm\frac{ia}{2}$

38. 1.

39. $2e.$

40. –1

41. $2\pi i.$

42. 0

43. $2\pi i.$

44. $-4\pi i.$

45. $-2\pi i.$

46. $\sum\left(\frac{1}{2^{n+1}}-1\right)(z+1)^n; |z+1| < 1$

47. $\sum(-1)^n\left\{\frac{2}{(2+i)^{n+1}}-\frac{1}{(1+i)^{n+1}}\right\}(z-i)^n.$

48. (i) $1+\sum(-1)^n\left\{\frac{3}{2^{n+1}}-\frac{8}{3^{n+1}}\right\}z^n.$

(ii) $1+3\sum(-1)^n\frac{2^n}{z^n+1}-8\sum(-1)^n\frac{z^n}{3^{n+1}}$

(iii) $1+\sum(-1)^n\left\{3.2^n-8.3^n\right\}\frac{1}{z^{n+1}}$

49. (i) $\sum\left(-\frac{1}{2.4^n}+\frac{1}{3^{n+1}}\right)(z+2)^n;$

(ii) $-\frac{1}{2}\sum\frac{(z+2)^n}{4^n}-\sum\frac{3^n}{(z+2)^{n+1}}$ (iii) $\sum\left(2.4^n-3^n\right)\frac{1}{(z+2)z^{n+1}}$

50. (i) $-\frac{3}{z+1}-\sum\left(1+\frac{2}{3^{n+1}}\right)(z+1)^n;0<|z+1|<1$

(ii) $-\frac{3}{z+1}+\sum\frac{1}{(z+1)^{n+1}}-2\sum\frac{(z+1)^n}{3^{n+1}};1<|z+1|<3.$

(iii) $-\frac{3}{z+1}+\sum\left(1+2.3^n\right)/z^{n+1},|z+1|>3.$ Res. at $(z=-1)=-3.$

51. (i) $1+\frac{4/5}{z-3}+\sum(-1)^n\left\{\frac{1}{2^{n+1}}+\frac{1}{5^{n+2}}\right\}(z-3)^n$, in $0<|z-3|<2$.

(ii) $1+\frac{4/5}{z-3}+\sum(-1)^n\left\{\frac{2^n}{(z-3)^{n+1}}+\frac{1}{5^{n+2}}.(z-3)^n\right\}$, in $2<|z-3|<5$

(iii) $1+\frac{4/5}{z-3}+\sum(-1)^n\left\{2^n+5^{n-1}\right\}\frac{1}{(z-3)^n}$, in $|z-3|>5$. Res. at $(z=3)=\frac{4}{5}$

52. (i) $-\frac{3}{z-1}-\frac{1}{(z-2)^2}-4\sum(z-1)^n$; Res. at $(z=1)=-3$; $0<|z-1|<1$

(ii) $\frac{4}{z-2}-\sum(-1)^n(n+4)(z-2)^n$; Res. at $(z=2)=4$; $0<|z-2|<1$

53. (i) $\frac{1}{z^2}-\frac{1}{2}\sum\left\{(-1)^n+1\right\}i^nz^n$, ; Res. at $(z=0)=0$.

(ii) $\frac{i/2}{z-i}-\sum\left\{(n+1)+\frac{1}{2^{n+2}}\right\}i^n\,(z-i)^n$; Res. at $(z=i)=\frac{i}{2}$.

(iii) $\frac{-i/2}{z+i}-\sum\left\{(n+1)+\frac{1}{2^{n+2}}\right\}(-1)^n(z+i)^n$; Res. at $(z=-i)=-\frac{i}{2}$.

54. (i) Res. at the simple pole $(z=-1+2i)=\frac{1}{4}(2+i)$

and that at $(z=-1-2i)=\frac{1}{4}(2-i)$

(ii) $z = 0$ is a double pole with Res. = –1/2; $z = -1 + i$ is a simple pole with Res. = 1/4; $z = -1 - i$ is a simple pole with Res. = 1/4.

55. (i) $z = 1$ is a simple pole; Res = 1/4; $z = -1$ is a double pole; Res. = –1/4

(ii) $z = -1$ is a double pole; Res $= -\frac{14}{25}$; $z = 2i$ is a simple pole with Res. $= \frac{1}{25}(7+i)$; $z = -2i$ is a simple pole with Res. $= \frac{1}{25}(7-i)$

56. (i) $z = 2e^{i\pi/4}, 2e^{i3\pi/4}, 2e^{i5\pi/4}, 2e^{i7\pi/4}$ are simple poles. Cor. residues are,

$$\frac{1}{32\sqrt{2}}(-1-i),\ \frac{1}{32\sqrt{2}}(1-i),\ \frac{1}{32\sqrt{2}}(1+i),\ \frac{1}{32\sqrt{2}}(-1+i).$$

(ii) $z = ae^{i\pi/4}, ae^{i3\pi/4}, ae^{i5\pi/4}, ae^{i7\pi/4}$ are simple poles. Cor. residues are

$$\frac{-i}{4a^2}, \frac{i}{4a^2}, \frac{-i}{4a^2}, \frac{i}{4a^2}.$$

57. (i) πi; (ii) $\frac{\pi i}{3}$;

58. $-2\pi i$; (ii) $-\frac{\pi^2}{2}i$

59. (i) $-2\pi i$; (ii) πi; (iii) $-\frac{\pi i}{4}$

60. (i) $\pi(i-2)$; (ii) $\frac{3\pi}{256}$; (iii) $\frac{-2\pi i}{9}$.

Exercise 4(c)

4. $\frac{\pi}{6}$
5. $\frac{\pi}{200}$
6. $\frac{5\pi}{12}$
7. $\frac{\pi}{2\sqrt{2}a^3}$
8. $\frac{\pi}{16}$
9. $\frac{\pi}{6}$
10. $\frac{\pi}{3}$
11. $\frac{\pi}{b}e^{-ab}$
12. $\frac{\pi}{30}\left(\frac{3}{e^2}-\frac{2}{e^3}\right)$
13. $\frac{\pi}{2}e^{-1/\sqrt{2}}\sin\left(\frac{1}{\sqrt{2}}\right)$
14. $\frac{\pi}{3e^2}(e-1)$
15. $\frac{\pi}{2e}$
16. $-\frac{\pi}{e}\sin 2$
17. $\frac{\pi}{2}e^{-2\pi}$
18. $\frac{\pi}{2}-\frac{3\pi}{4e}$
19. $\frac{2\pi}{1-p^2}$
20. $\frac{2\pi}{\sqrt{a^2-b^2}}$
21. $\frac{2\pi}{b^2}\left\{a-\sqrt{a^2-b^2}\right\}$
22. $\frac{2\pi}{\sqrt{1+a^2}}$
23. $\frac{\pi}{6}$
24. $\frac{\pi}{12}$
25. $\frac{3\pi}{8}$